



## Infection Prevention and Control Manual

# Standard and Transmission-based Precautions Policy

### Scope

| Site      | Service/Department/Unit | Disciplines |
|-----------|-------------------------|-------------|
| SCGHOPHCG | Organisation Wide       | All Staff   |

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## Introduction

Standard and transmission-based precautions describe specific work practices required to reduce the risk of transmission of infectious agents in healthcare.<sup>1</sup>

Standard precautions is the primary strategy for minimising the transmission of infection and must be applied to all patients.<sup>1</sup>

Transmission-based precautions are used **in addition** to standard precautions, where transmission is not completely interrupted by using standard precautions alone and when patients are suspected or confirmed to be infected with infectious agents transmitted by the contact, droplet, or airborne routes.<sup>1</sup>

**NOTE: for COVID-19 refer to SCGOPHCG COVID-19 Policy.**

## Risk Statement

Non-compliance with this policy will:

|                                       |                                     |                                   |                                     |
|---------------------------------------|-------------------------------------|-----------------------------------|-------------------------------------|
| Breach legislative requirements       | <input checked="" type="checkbox"/> | Impact on Patient Quality of Care | <input checked="" type="checkbox"/> |
| Breach National/State/Hospital Policy | <input checked="" type="checkbox"/> | Impact on Patient Safety          | <input checked="" type="checkbox"/> |
| Breach professional standards         | <input checked="" type="checkbox"/> | Misconduct                        | <input type="checkbox"/>            |
| Breach SCGOPHCG Mission & Values      | <input checked="" type="checkbox"/> | Staff Safety                      | <input checked="" type="checkbox"/> |
| Other:                                | <input type="checkbox"/>            |                                   |                                     |

## Definitions

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Aerosol-generating procedure (AGP)</b> | Are those procedures that promote the generation of fine airborne particles (aerosols) that may result in the risk of airborne transmission of disease. <sup>1,2</sup> Refer to Appendix 2 for more details.                                                                                                                                                                                                            |
| <b>Blood and body substances</b>          | Refers to blood, including dried blood, and all other body substances, secretions and excretions, including saliva, urine and faeces, semen, vaginal secretions, cerebrospinal fluids, synovial fluid, pleural fluid, pericardial fluid, peritoneal fluid, amniotic fluid (but excluding sweat), regardless of whether they contain visible blood. <sup>1</sup>                                                         |
| <b>Cohorting</b>                          | Placing patients together in the same room who are infected with the same pathogen.                                                                                                                                                                                                                                                                                                                                     |
| <b>Donning</b>                            | To put on or apply PPE.                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Doffing</b>                            | To take off or remove PPE.                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Fit checking</b>                       | The appropriate minimum standard at the point of use for staff members using P2 or N95 respirators. No clinical activity shall be undertaken until a satisfactory fit has been achieved. It involves a check each time the respirator is put on to ensure the respirator is properly applied, that a good seal is achieved over the bridge of the nose and mouth and there are no gaps between the respirator and face. |
| <b>Fit Test</b>                           | Use of a qualitative or quantitative method to measure the quality of the seal between the respirator and the wearer's face to ensure that a respirator is fitted correctly and suitable for use by the individual. <sup>3</sup>                                                                                                                                                                                        |

|                                                              |                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P2 or N95<br/>Particulate filter<br/>respirator (PFR)</b> | Respirators that meet the required standards for filtering efficiency. P2: meets Australia/European standards at 94%. <sup>4</sup> N95 meets USA Standards at 95% efficiency. <sup>5</sup> Both P2 and N95 respirators are considered appropriate for use with airborne precautions. |
| <b>Personal<br/>Protective<br/>Equipment (PPE)</b>           | A variety of barriers used alone or in combination to protect mucous membranes, skin and clothing from contact with infectious agents. PPE may include gloves, gowns, masks, respirators, protective eyewear and, face shields.                                                      |
| <b>Source control</b>                                        | The use of different measures applied to individuals to contain an infectious agent at either the portal of exit from the body or within a confined space. <sup>6</sup>                                                                                                              |

## Policy Principles

The purpose of this policy is to provide details to all staff on the application of standard and transmission-based precautions to prevent and control infection transmission within healthcare. All staff have a responsibility to adhere to standard and transmission-based precautions at all times.

### 1. Standard Precautions

Standard precautions are primary work practices implemented to reduce the risk of transmission of infectious agents to and from healthcare workers (HCWs) and patients and the subsequent risk of healthcare-associated infections. Standard precautions apply to all patients regardless of known diagnosis or infectious status and must be used when handling blood and body substances, secretions and excretions (excluding sweat), regardless of whether they contain visible blood, non-intact skin and mucous membranes.<sup>1</sup>

Standard precautions consist of:

- Hand hygiene in accordance with the 5 moments for hand hygiene<sup>7</sup>
- Use of appropriate personal protective equipment (PPE) – this includes gloves, plastic aprons, long sleeved fluid resistant gowns, masks, protective eyewear when there is a likelihood of contact with blood, other body fluids, secretions and excretions (except sweat), non-intact skin and mucous membranes. Refer to Appendix 1 for further information
- Aseptic technique to minimise contamination and protect the patient from infection during procedures
- Environmental cleaning and management of spills
- Reprocessing of re-useable medical equipment and instruments
- Respiratory hygiene and cough etiquette - covering sneezes and coughs with a disposable tissue helps prevent infected persons from dispersing respiratory secretions into the air and should be applied at all times.<sup>1</sup>
- Appropriate handling of linen and waste
- Safe use and disposal of sharps.

In addition, HCWs should be immunised against vaccine-preventable diseases – refer to SCGOPHCG Healthcare Worker Health and Immunisation Policy.

For further information Refer to [‘Related Documents’](#) to access the relevant SCGOPHCG Policies.

## 2. Transmission-based precautions

Transmission-based precautions are used **in addition** to standard precautions when patients are suspected or confirmed to be infected with agents where transmission is not completely interrupted using standard precautions alone.<sup>1</sup>

Based on the routes of disease transmission, there are three types of transmission-based precautions:

- contact precautions
- droplet precautions
- airborne precautions

Some diseases can have more than one route of transmission e.g. varicella (chickenpox). Early initiation of transmission-based precautions is essential in reducing further transmission opportunities.<sup>1</sup>

Refer to SCGOPHCG IP&C Manual: Infectious Disease Summary Table Guideline to determine the type of transmission-based precautions required.

Staff information sheets on specific organism transmission-based precaution requirements can be downloaded from HealthPoint/Infection Prevention and Control/Staff Information Sheets page.

### 2.1 Minimum requirements for transmission-based precautions

For COVID-19 – follow the SCGOPHCG COVID-19 policy.

The following minimum measures are recommended for **all patients** requiring transmission-based precautions:

- Implementation of standard precautions at all times
- Patients must have an appropriate colour coded precaution sign displayed outside their room
- Allocate a single room with ensuite facilities
- Explain the rationale of the required precautions to patient and family. The patient and family can be provided with an information brochure called 'Why am I in Isolation' and can be downloaded from HealthPoint/Infection Prevention and Control/Patient Information Sheets. The potential for anxiety, mood disturbances, perceptions of stigma and reduced contact with clinical staff must be considered for all patients when transmission-based precautions are applied. Clear explanations and rationales should be provided to patients to alleviate these affects.<sup>1</sup> Patient information sheets on specific organisms can be downloaded from HealthPoint/Infection Prevention and Control/Patient Information Sheets.
- Staff must select and use appropriate PPE depending on the required transmission-based precaution and risk assessment<sup>1</sup> - refer to Appendix 1
- Essential staff only to enter transmission-based precaution rooms
- Maintain notes and bedside folders outside of room
- Equipment
  - Keep equipment and consumables in the room to a minimum
  - Consumables that have been opened or cannot be decontaminated are to be discarded on patient discharge
  - Dedicate non-critical items to the patient room e.g. stethoscope, tourniquet or use single patient disposable items
  - Decontaminate equipment using a combined detergent/disinfectant wipe and allow to dry before use on another patient
- Enhanced cleaning and disinfection of the patient environment may be initiated
- Patient's medication to be stored in the medication room or medication cabinet outside of patient room

- Restrict transfer of patients within and between facilities. If transfer required, the reason for transmission-based precautions must be handed over to the receiving department/facility, and where relevant, transport services, prior to patient transfer
- Patient cohorting is only to occur on advice from the Infection Prevention and Control Unit
- Visitors – in most instances patients on transmission-based precautions may receive visitors. However, where possible limit the number of visitors. Babies and toddlers should be discouraged from visiting. In some instances, there may be additional restrictions or visitors may be required to wear PPE. Staff will need to provide education and supervision for the donning and doffing of PPE and hand hygiene. For specific restrictions refer to the staff information sheets on specific organisms available on the Infection Prevention and Control page on HealthPoint.

## 2.2 Contact Precautions

Contact transmission is the most common mode of transmission and may be direct or indirect:

- **Direct contact** – occurs when infectious agents are transferred directly from one person to another person<sup>1</sup>
- **Indirect contact** – involves the transfer of an infectious agent through a contaminated object (fomite) or person e.g. HCWs hands transmitting infectious agents and not performing proper hand hygiene, or contact with fomites (e.g. bedding, stethoscopes).<sup>1</sup>

In **addition** to standard precautions, and the minimum requirements for transmission-based precautions outlined in Section 2.1 contact precautions include:

- Patient placement in a single room with ensuite facilities
- Orange '**contact precaution**' sign to be placed outside the patient's single room. Door may be left open unless advised by infection Prevention and Control
- Donning appropriate PPE including gloves, apron or gown. Facial protection may be required as for standard precautions. Refer to Appendix 1 and Appendix 3
- Appropriate doffing procedures, including hand hygiene immediately on exit of the patient's room.<sup>1</sup> Refer to Appendix 4
- Transfer to other departments:
  - Inform the transporting staff and receiving department prior to transfer<sup>1</sup>
  - Transport staff do not routinely need to wear PPE, unless required for standard precautions
  - Hand hygiene must be performed after transporting the patient.

## 2.3 Droplet Precautions

Respiratory droplets are generated from persons coughing, sneezing or talking and during certain procedures. Droplet transmission occurs when these droplets, carrying infectious pathogens, travel directly from the respiratory tract of the infected individual to a susceptible mucosal surface i.e., conjunctivae, nasal mucosa or mouth.<sup>1</sup>

Close proximity (less than 1.5 metres) is required for transmission, as the droplets do not remain suspended in the air and generally only travel short distances.<sup>1</sup> Droplets can contaminate horizontal surfaces close to the source patient with contact transmission occurring, usually via HCW hands contaminated on contact with those surfaces.<sup>1</sup>

In **addition** to standard precautions, and the minimum requirements for transmission-based precautions outlined in Section 2.1 droplet precautions include:

- Patient placement in a single room with ensuite facilities - door must be kept closed.  
**Note:** the door can be open if the patient is ventilated on a closed circuit except during aerosol generating procedures.

In the case of a patient demonstrating challenging behaviours, making in room isolation difficult, in addition to utilising strategies to support mental health, consider the suitability of 1:1 monitoring

- Green '**droplet precaution**' sign to be placed on the door outside the patient room.
- Donning of appropriate PPE including a fluid resistant surgical mask (minimum level 2) protective eyewear/full face visor, gloves, and long sleeved fluid resistant gown prior to entry to the room.<sup>1</sup> Refer to Appendix 1 and Appendix 3
- Appropriate doffing procedures, including hand hygiene immediately on exit of the patient's room.<sup>1</sup> Refer to Appendix 4
- Patients required to undergo an aerosol generating procedure (AGP):
  - Airborne PPE (a *fit checked*) P2 or N95 respirator in addition to gown, gloves and protective eyewear) to be worn for the period of the procedure and for at least 30 minutes following procedure if staying in the room – refer to Appendix 1
  - Door to room to remain closed for the period of the procedure and for at least 30 minutes following procedure
  - Limit the numbers of HCWs in the room when AGPs are performed.<sup>3</sup>
- The importance of respiratory hygiene and cough etiquette is to be emphasised with the patient.<sup>1</sup>
- Transfer to other departments:
  - Patients should wear a surgical mask during transport
  - Transport staff do not routinely need to wear droplet PPE unless the patient cannot wear a mask e.g. on BIPAP or as otherwise indicated by Infection Prevention and Control for specific infections e.g. COVID-19
  - **Ensure no other staff or visitors are in the lift during transfer**
  - For patients with Influenza-like Illness (ILI) or COVID-19 – refer to SCGOPHCG IP&C Manual: ILI policy and COVID-19 Policy.

## 2.4 Airborne Precautions

Diseases transmitted via the airborne route occurs by dissemination of airborne droplet nuclei or small particles in the respirable size range (5 µm or smaller) containing infectious agents that can remain suspended in the air for long periods and infective over time and distance. These small particles can transmit infection into small airways of the respiratory tract.<sup>1</sup>

In **addition** to standard precautions, and the required minimum transmission-based precautions as outlined in Section 2.1, **airborne precautions** includes:

- Patient to be placed in a Negative Pressure Isolation Room (NPIR) as this has special ventilation requirements. Door must be kept closed and negative pressure activated.  
**NOTE:** There are no NPIRs located at OPH.
- Blue '**airborne** precaution' sign to be placed outside the patient's room.
- Donning of appropriate PPE including *fit checked* P2 or N95 respirator, protective eyewear/full face visor, long sleeved fluid resistant gown and gloves prior to entry to the room<sup>1</sup> – refer to Appendix 1. If the patient is in an NPIR, then donning to occur in the anteroom<sup>1</sup>
- Appropriate doffing procedures as per Appendix 4 is to be followed immediately on exit of the patient's room. If the patient is in an NPIR, then doffing to occur in the anteroom.
- Transfer of the patient outside the NPIR:
  - Inform transport staff and receiving department prior to transfer
  - Patient should wear a surgical mask during transportation or when outside the NPIR at all times<sup>1</sup>
  - Transport staff do not routinely need to wear airborne PPE, unless the patient cannot wear a mask e.g. on BIPAP or as otherwise indicated by Infection Prevention and Control for specific infections e.g. COVID-19
  - Ensure no other staff or visitors are in the lift

- Cover any lesions associated with the condition (e.g. chickenpox)<sup>1</sup>
- For patients with COVID-19 refer to SCOPHCG COVID-19 Policy

**NOTE:** HCWs, including students who are required to wear a P2 or N95 respirator **must** be fit tested for P2 or N95 respirator and **must** undertake a competency assessment for donning, fit checking and doffing of a P2 or N95 respirator.

### ***Negative Pressure Isolation Room (NPIR)***

SCGH Negative Pressure Isolation Rooms are located on:

- Ward G54 x 2 – rooms 9 and 10
- Ward G71 x 1 – room 15 and room 14 used for the anteroom
- Intensive Care Unit x 2 – N9 and N10
- Emergency Department x 2

**Note:** OPH does not have Negative Pressure Isolation Rooms.

OPH patients requiring airborne precautions must be nursed in a single room with door closed. A risk assessment is to be undertaken between the medical team and Infectious Diseases Consultant to determine if transfer arrangements are to be made for SCGH.

NPIRs must be used for patients with the following conditions:

- Pulmonary tuberculosis
- Chickenpox or other disseminated varicella zoster virus infections
- Measles
- Respiratory Viruses (non-endemic) e.g. Avian influenza, Severe Acute Respiratory Syndrome (SARS), Middle East Respiratory Syndrome Coronavirus (MERS-CoV).
- Viral Haemorrhagic Fever
- COVID-19 (Refer to SCOPHCG COVID-19 Policy).

### ***Operation of NPIRs***

The NPIR should be put into 'isolation mode' (negative pressure) prior to patient placement:

- Close the doors
- Turn the key outside room to "on"
- When the room is functioning at negative pressure the 'isolation mode' light will illuminate
- When in 'isolation mode' the doors must not be left open
- The pressure differential should be visually checked prior to patient admission to the room at least daily by clinical staff.

### ***Reporting faults***

- NPIRs are monitored by the Building Management System (BMS) with alarm parameters set in place for negative pressure. The BMS automatically tracks and monitors room pressures and printouts are available to check room compliance.
- When the rooms are in ISOLATION MODE, if any faults are identified (e.g. isolation mode light goes off, doors not closing) report immediately to the Shift Engineer.
- Inform the CNS of the area.

## Compliance, monitoring and evaluation.

Managers and Heads of Departments are required to ensure all staff have completed the required mandatory Infection Prevention and Control education including:

- Annual Hand Hygiene e-learning
- Annual Infection Prevention and Control e-learning
- Aseptic technique e-learning and competency assessment

The Infection Prevention and Control Unit, together with Clinical Nurse Educators, review and participate in audits for PPE compliance with donning and doffing procedures.

Environmental hygiene is assessed in collaboration with the Infection Prevention and Control Unit and Patient Support Services Management.

All NPIRs undergo regular testing and maintenance as per NMHS Facility Management maintenance schedules.

## Related Documents

- [OPH, Waste Management Manual](#)
- [NMHS, OSH Personal Protective Equipment \(PPE\) Policy](#)
- [SCGH Hospital Policy Waste and Environmental Management #079](#)
- [SCGOPHCG, IP&C policies](#)
  - Prevention and Management of Sharps Injuries and Body Substance Exposure Policy
  - Cleaning of Patient Care Equipment
  - COVID-19
  - Healthcare Worker Health and Immunisation
  - Linen Management
  - Environmental Cleaning
  - Meningococcal Disease (Invasive)
  - Influenza-Like Illness
  - Tuberculosis
  - Multi-Resistant Organisms
  - Respiratory Viruses (Non-Endemic)
  - Measles
  - *Clostridioides Difficile*

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## Authorisation & Version Control

|                       |                                                                                                                                                                                 |                |              |              |              |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|--------------|--------------|
| Policy Sponsor        | Nurse Co-Director, Specialty and Ambulatory Services                                                                                                                            |                |              |              |              |
| Policy Contact        | Coordinator of Nursing, Infection, Prevention and Control                                                                                                                       |                |              |              |              |
| Date First Issued:    | 01/08/1986                                                                                                                                                                      | Last Reviewed: | October 2021 | Review Date: | October 2024 |
| Version No.           | One                                                                                                                                                                             |                |              |              |              |
| Endorsed by:          | SCGOPHCG Infection, Prevention and Control Committee                                                                                                                            |                |              | Date:        | August 2021  |
| Standards Applicable: | NSQHS Standard 3 (Version 2):<br>Preventing and controlling Healthcare-associate infection.  |                |              |              |              |

For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

This document can be made available in alternative formats on request for a person with a disability.

## Appendix 1: Guidance on the use of PPE

### Personal protective equipment (PPE)

PPE refers to a variety of barriers, used alone or in combination to protect mucous membranes, airways, skin and clothing from contact with infectious agents.<sup>1</sup>

PPE must be readily available and accessible. PPE is designed for a purpose and should not be worn outside the immediate area in which it is used.

The selection of PPE should be based on the risk of transmission of potentially infectious microorganisms to the worker from:

- exposure to blood and body substances during activity (standard precautions)
- contamination from infectious microorganisms via the contact, droplet or airborne route (transmission-based precautions)<sup>10</sup>

All staff **must** be compliant with their mandatory annual PPE training and deemed competent in wearing and performing a fit check for P2 or N95 respirators. Only compliant staff should enter the room of a patient being managed on transmission-based precautions.

### Gloves

Gloves are worn to protect healthcare worker hands when:

- anticipating direct contact with blood or body substances, mucous membranes, non-intact skin and other potentially infectious material, including during any invasive procedure
- handling or touching visibly or potentially contaminated patient-care equipment and environmental surfaces
- having direct contact with patients who are colonised or infected with pathogens transmitted by the contact route e.g. according to recommendations for contact precautions.<sup>1</sup>

Appropriate use of gloves includes;

- Performing hand hygiene before donning and following doffing of gloves
- Changing gloves and performing hand hygiene in accordance with the 5 Moments for hand hygiene, this includes:
  - between episodes of care for different patients
  - between each episode of clinical care on the same patient to prevent cross contamination of body sites e.g., mouth care followed by wound care
  - when the integrity of the glove has been compromised or contaminated e.g. ripped, torn or visibly soiled
- Gloves are single use items and must not be washed. Alcohol-based hand rub must not be applied to gloved hands<sup>1,7</sup>
- When removing gloves, care should be taken not to contaminate hands.<sup>1</sup>

**Wearing gloves does not remove the need for hand hygiene.<sup>7</sup>**

### Plastic aprons/long sleeved fluid resistant gowns

Disposable plastic aprons/disposable long sleeved gowns are single use and are to be worn by HCWs:

- when there is a risk of contamination/contact with blood or body substances, mucosal surfaces and/or non-intact skin (standard precautions).
- to prevent transmission of an infectious agent to the HCW skin via direct transfer from the patient or inadvertent contact and indirect transmission from the contaminated environment (transmission-based precautions).<sup>1</sup>

The type of apron/gown selected will be determined by:

- the nature of patient interaction
- anticipated degree of contact with infectious material and the potential for blood or body substances to penetrate through to clothes or skin.

Disposable plastic aprons are recommended for general use when there is a low to moderate risk of sprays or spills or exposure to blood or body substances during low risk procedures e.g. empty drain/catheter, wound care or direct physical contact is minimal.

Plastic aprons/disposable long sleeved gowns are to be removed and hand hygiene performed **before** leaving the patient-care area (e.g. in the room, immediately on exit of the room or in the anteroom), in a manner that minimises the risk of person, other or environmental contamination.

### Respiratory protection

For use of face masks and respirators in the context of COVID-19 please follow the SCGOPHCG COVID-19 Policy.

During periods of mandatory mask wearing please refer to the “Wearing of Face Masks During Mandated Periods’ fact sheet found on the [COVID-19 Information for Staff and Resources page](#)

### Surgical face masks

Surgical masks create a physical barrier to the mouth, nose and respiratory tract from large splashes and droplets.<sup>1</sup> They do not provide protection against airborne particles.

Surgical masks used by HCWs for protection against microorganisms must be disposable and fluid resistant.

Surgical masks have different properties and are rated as level 1, 2 or 3 according to the degree of barrier protection, refer to Table 1.

**Table 1: Levels of barrier protection and application of surgical masks**

| Characteristics                                                                                          | Level 1 barrier                                                                                                                                                          | Level 2 barrier                                                                                                                                     | Level 3 barrier                                                                                                            |
|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Application                                                                                              | For general medical procedures where the wearer is not at risk of blood or body fluid splash, or to protect staff and/or patient from droplet exposure to microorganisms | For use in emergency departments, dentistry, changing dressings on small wounds or healing wounds where minimal blood or droplet exposure may occur | For all surgical procedures, major trauma first aid or in any areas where the HCW is at risk of blood or body fluid splash |
| Bacterial Filtration Efficiency (BFE) %                                                                  | ≥95                                                                                                                                                                      | ≥98                                                                                                                                                 | ≥98                                                                                                                        |
| Differential pressure (ΔP), mm H <sub>2</sub> O/cm <sup>2</sup>                                          | <4.0                                                                                                                                                                     | <5.0                                                                                                                                                | <5.0                                                                                                                       |
| Resistance to penetration by synthetic blood (fluid resistance) minimum pressure in mmHg for pass result | 80 mmHg                                                                                                                                                                  | 120 mmHg                                                                                                                                            | 160 mmHg                                                                                                                   |

*Extract from AS 4381:2015 Single use surgical face mask use in healthcare*

When wearing a surgical mask the following applies;

- Masks must be donned and doffed in a manner to prevent contamination to the wearer
- Masks should be changed between patients
- Inspect the mask for tears or holes prior to donning. Do not use if damaged or soiled
- Masks are to be changed four hourly at a minimum, if contaminated with blood or body fluids, become moist or difficult to breathe through
- Masks are single use and disposable and should never be reapplied after they have been removed
- Do not pull the mask below the nose or chin or hang around the neck
- Do not touch or adjust the front of the mask or face as this may contaminate the wearer. If accidentally touched, the HCW must remove themselves from the patient room to replace the mask, perform hand hygiene before and after removal of the soiled mask. **Note:** does not apply to the operating theatre setting.

### Respiratory protection - P2 or N95 respirator

Particulate filter respirators (PFRs), such as a P2 or N95 or equivalent are used to protect the wearer from exposure to small airborne particles.<sup>9</sup> PFRs must meet the requirements of AS/NZS1716:2012.

A correctly fitted P2 or N95 respirator must be worn when:

- Undertaking routine care of patients with confirmed or suspected diseases transmitted by the airborne route e.g. measles, chickenpox, active pulmonary tuberculosis
- Caring for patients with confirmed or suspect COVID-19-refer to the COVID-19 policy
- When performing aerosol generating procedures (AGPs), on patients with a known or suspected respiratory infection or other disease transmitted via the droplet or airborne route. This includes staff who may enter or are in the room where the AGP is undertaken either during the procedure or 30 minutes following the procedure/patient departure from the room.<sup>1</sup> Refer to Appendix 2 for information relating to AGPs.
- For staff performing and present during high-risk aerosol generating procedures i.e. bronchoscopy, bronchoalveolar lavage and induced sputum specimen collection.

**Note:** Some brands of respirator have lower fluid resistance and must be worn with a full face shield. Refer to the WA Department of Health [‘N95/P2 Respirator Options’](#) information sheets.

When wearing a P2 or N95 respirator the following applies:

- The front of the respirator is considered soiled. Do not touch or adjust the respirator once inside the patient’s room.<sup>1</sup>
- Respirators should be changed when they become moist or the wearer is having difficulty breathing<sup>1</sup>
- Respirators are single use and disposable and should never be reapplied after they have been removed<sup>1,6</sup>
- Respirators must not be left dangling around the neck<sup>1</sup>
- Hand hygiene should be performed prior to donning, upon touching or discarding a used respirator
- All HCWs wearing a P2 or N95 respirator **must** be fit tested. Staff are required to know the respirator brand, style and size that they have passed a fit test for and ensure this is available for use in their area of work
- Staff who are unable to achieve the correct fit (e.g. due to facial hair or other problems with fitting the respirator) **must** not enter the room
- Safe donning and doffing procedures are an essential component in preventing healthcare worker exposure.<sup>1</sup> Refer to appendix 3 and 4 for correct donning and doffing procedures.

### Facial fit

Facial fit is a prime factor in obtaining good protection when using respiratory protective equipment. Achievement of an effective seal between the wearers face and respirator prevents inward leakage of contaminants.<sup>2</sup> The respirator must be a suitable size for the persons face.<sup>6</sup>

Facial hair and facial piercings must not interfere with the safe use and seal of the respirator. Staff who have facial hair (including 1-2 days growth/stubble) must be aware that an adequate seal cannot be guaranteed between the respirator and the wearers face.

Employees must present to work prepared to wear PPE as required for their role - For further information refer to [‘Related Documents’](#) to access the relevant NMHS OSH Policies

### Fit testing

Fit testing is a validated process utilising specialised equipment that identifies the optimal respirator size and style for an individual.<sup>2</sup> The NMHS Occupational Health and Safety Department co-ordinates the Fit testing program for SCGOPHCG staff.

Fit testing does not guarantee that a respirator will not leak and does not replace the need for fit checking to ensure a good seal, each time a respirator is used.<sup>2</sup>

### Fit-checking

Fit checking describes the process that all staff must perform each and every time a P2 or N95 respirator is worn to ensure a good facial seal is achieved i.e. the respirator is sealed over the bridge of the nose and mouth and that there are no gaps between the respirator and face when worn. Fit-checking is the most reliable way to ensure an adequate facial seal on each occasion. No clinical activity is to be undertaken until a satisfactory fit has been achieved.<sup>1,2</sup>

Staff who are unable to achieve the correct fit (e.g. due to facial hair or other problems with fitting the respirator) should not enter the room. Healthcare workers with facial hair, including 1-2 days of beard growth, cannot be guaranteed adequate seal between the P2 or N95 respirator and the face.<sup>1</sup>

HCWs required to wear a P2 or N95 respirator **must** have been fit tested for P2 or N95 respirator mask and **must** undertake a competency assessment for donning, fit checking and doffing of a P2 or N95 respirator mask including;

- staff who have contact with patients on transmission-based precautions
- staff who perform and assist with bronchoscopy procedures
- staff performing induced sputum procedures.

Refer to HealthPoint/Infection prevention and Control Education/Other resources – [SCGOPHCG Personal Protective Equipment – Respiratory Protection Competency](#)

### Protective eyewear - goggles/face visors/shields

Protective eyewear such as face shields, visors, goggles and wrap around safety glasses provides close fitting eye protection from splashes, sprays and respiratory excretions. Prescription glasses are not adequate to protect the wearer from splashes or for respiratory protection.<sup>9</sup>

Re-usable protective eyewear i.e. goggles, should be cleaned with a combined detergent/disinfectant wipe before being stored dry in a clean location.<sup>1</sup> Hand hygiene must be performed prior to and following removal of protective eyewear.<sup>1</sup>

The front of eyewear is considered contaminated. Items must be removed using the ties, earpieces and/or headband used to secure equipment to the head.<sup>1</sup>

**Note:** Appropriate protective eyewear must be available to fit over personal eyeglasses<sup>1</sup>

**Table 2: Minimum PPE requirements for transmission-based precautions**

| Type of precautions  | Gloves                                                   | Apron/Gown    | Mask/respirator                             | Eye protection |
|----------------------|----------------------------------------------------------|---------------|---------------------------------------------|----------------|
| Standard             | PPE is worn to prevent exposure to blood and body fluids |               |                                             |                |
| Contact              | ✓                                                        | Apron or Gown | If required as part of standard precautions |                |
| Droplet              | ✓                                                        | Gown          | Surgical                                    | ✓              |
| Airborne             | ✓                                                        | Gown          | P2/N95*                                     | ✓              |
| Contact and droplet  | ✓                                                        | Gown          | Surgical                                    | ✓              |
| Contact and airborne | ✓                                                        | Gown          | P2/N95*                                     | ✓              |

\* Some brands of respirator have lower fluid resistance and must be worn with a full-face shield

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## Appendix 2: Aerosol-generating procedures (AGPs)

Aerosol generating procedures (AGPs) are those that stimulate coughing and/or sneezing and promote the generation of fine airborne particles (aerosols) resulting in the risk of airborne transmission.<sup>1</sup>

**AGPs must not be undertaken on patients with suspected or confirmed Influenza Like Illness (ILI) or COVID-19 unless strictly necessary and authorised by the Medical Consultant.**

The following examples are illustrative of a range of AGPs<sup>2</sup>

- Insertion or removal of endotracheal tube
- Intentional or inadvertent disconnection/reconnection of closed ventilator circuit
- High frequency oscillatory ventilation (HFOV)
- Open oropharyngeal or tracheal suctioning
- Upper respiratory instrumentation or surgery e.g. bronchoscopy, tracheotomy, ear nose throat surgery
- Surgical or post mortem procedures on respiratory tract involving high speed devices
- Intercostal catheter insertion for relief of pneumothorax
- Thoracic surgery that involves entering the lungs.

Other procedures that can generate respiratory aerosols

- Manual or on-invasive ventilation (NIV)
  - Bi-level positive airway pressure ventilation (BiPAP)
  - Continuous positive airway pressure ventilation (CPAP)
- Collection of induced sputum
- High flow nasal oxygen therapy
- Diagnostic instrumentation of the upper digestive tract including transoesophageal echocardiography
- Airway manoeuvres (e.g. suction, ventilation) during cardiopulmonary resuscitation (CPR).

## Appendix 3: Sequence for donning PPE



Government of Western Australia  
North Metropolitan Health Service  
Sir Charles Gairdner Osborne Park Health Care Group

Prepared by SCGOPHCG Infection Prevention and Control Unit February 2021

## Personal Protective Equipment (PPE)

# Sequence for Donning PPE

### 1 Perform Hand Hygiene

### 2 GOWN

- Select a long sleeved disposable gown
- The gown must completely cover the torso from neck to knees, arms to end of wrists, and wrap around the back
- Secure ties at back of neck and side of waist



### 3 FLUID REPELLENT MASK OR P2/N95 RESPIRATOR

- Choose appropriate mask / respirator for the type of precautions
- If P2/N95 worn ensure respirator is the correct size for your face
- Place over nose, mouth and chin
- Secure on head with ties or elastic bands
- Fit the flexible band over nose bridge
- Fit snug to face and below chin
- Do not touch front of mask or re-adjust mask once in patient's room
- If mask is damaged, soiled or damp, leave room, perform hand hygiene and change mask

**\*P2/N95 respirator key points;**

- Perform fit-check each time a new respirator is worn.



### 4 GOGGLES OR FULL FACE VISOR

- Place protective eye wear over your eyes, (and over prescription glasses if worn) and adjust to fit comfortably and securely

or

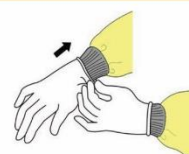
- Place visor over face and adjust to fit comfortably and securely



### 5 Perform Hand Hygiene

### 6 GLOVES

- Select correct glove type and size
- Extend gloves to cover gown cuffs



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## Appendix 4: Sequence and procedures for doffing PPE



Government of Western Australia  
North Metropolitan Health Service  
Sir Charles Gairdner Osborne Park Health Care Group

Prepared by SCGOPHCG Infection Prevention and Control Unit February 2021

## Personal Protective Equipment (PPE) Sequence for Doffing PPE

### 1

#### GLOVES

- The outside surface of gloves are considered contaminated
- Grasp outside of glove with opposite gloved hand; peel off
- Put removed glove in gloved hand
- Slide fingers of ungloved hand under remaining glove at wrist.
- Peel glove off from the inside over first glove
- Discard gloves into waste bin



### 2

#### Perform Hand Hygiene

### 3

#### GOWN

- Gown front and sleeves are considered contaminated
- Unfasten ties (top then waist)
- Peel gown away from neck and shoulders, touching inside of gown only
- Turn contaminated outside of the gown toward the inside
- Fold or roll into a bundle
- Discard gown into waste bin



### 4

#### Perform Hand Hygiene

### 5

#### GOGGLES OR FACE VISOR

- The outside of goggles/face visor are considered contaminated.
  - To remove, use only handle by head band or ear pieces
  - Reusable goggles\* to be cleaned after use with a combined detergent/disinfectant wipe
  - Discard visor into waste bin
- \*Reusable goggles key points;**
- May be reworn by an individual staff member after cleaning and disinfection
  - Clean and disinfect all surfaces after use e.g. inner surface first then outer surface
  - Store in a designated clean area until next use



### 6

#### Perform Hand Hygiene

### 7

#### MASK/RESPIRATOR

- Front of mask/respirator is contaminated – DO NOT TOUCH
- Grasp the bottom, then top tie/elastics and remove respirator
- Undo bottom ties then top ties and remove mask.
- Discard mask/respirator into waste bin



### 8

#### Perform Hand Hygiene